### Open loop system

When a NC system does not have any feedback arrangement, it is known as 'open look system'. In this system actual output movement of spindle and table is not compared with the desired movement given by input media.

### Closed loop system

The movements of spindle and sliders are measured and compared with the values given by input media. If the output values are not matched with input values, the difference is corrected in machine control unit of NC machine. This system is used where the higher accuracy is required.

## 9.11 Advantages of CNC machines over conventional machine tools

1. The process of production, planning becomes easy.
2. The production of a single CNC machine is equivalent to production of many conventional machine tools. So the time is lesser.
3. As a single CNC machine is doing many machine functions, it is not necessary to buy many machines. It ensures reduction of floor space.
4. The time required for making input media is very less, when compared with making jigs and fixtures.
5. Smaller quantities of production is possible so keeping low raw material is enough
6. The design change of the component can be done easily and quickly
7. As the CNC machines require very little attention of the operator, the machine utilization is better
8. Jigs and fixtures are not used in CNC machines. It reduces the tooling costs.
9. In a single set up of tool and work piece, many operations are done. So the time is saved
10. Due to higher accuracy, it leads to better assembly and reduces fitting costs. The quality of the product also becomes better.
11. Operators errors are less, so wastage is also less
12. As a single operator can supervise several machines at a time, the labour cost is reduced.

## 9.12 Programming for CNC Machines

Based on the component drawing, a lot of information are need to be fed into the control unit for processing and machining. The informations include dimensions, shape, cutting speed, feed, depth of cut, sequence of operations, tool and work material. The said information should be fed into the machine control unit in an acceptable form. Preparation of the instructions in the above form is known as 'programming'.

### Part Program

A series of instructions describing the part to be produced in an acceptable form to the control unit or computer is known as part program. Generally part programming is done be the following methods.

1. Manual assisted part programming
2. Computer assisted part programming

APT (Automatically programmed tools), ADAPT, EXAPT, SPLIT, PROMPT are some of the high level languages for writing NC programs.



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